# Third Rail Defect Detection System

## Image Processing Based Solution

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# Agenda


1. Executive Summary
2. Technical Solution
3. System Features
4. System Features
5. Performance & Results
6. Implementation
7. Conclusion

# Project Overview

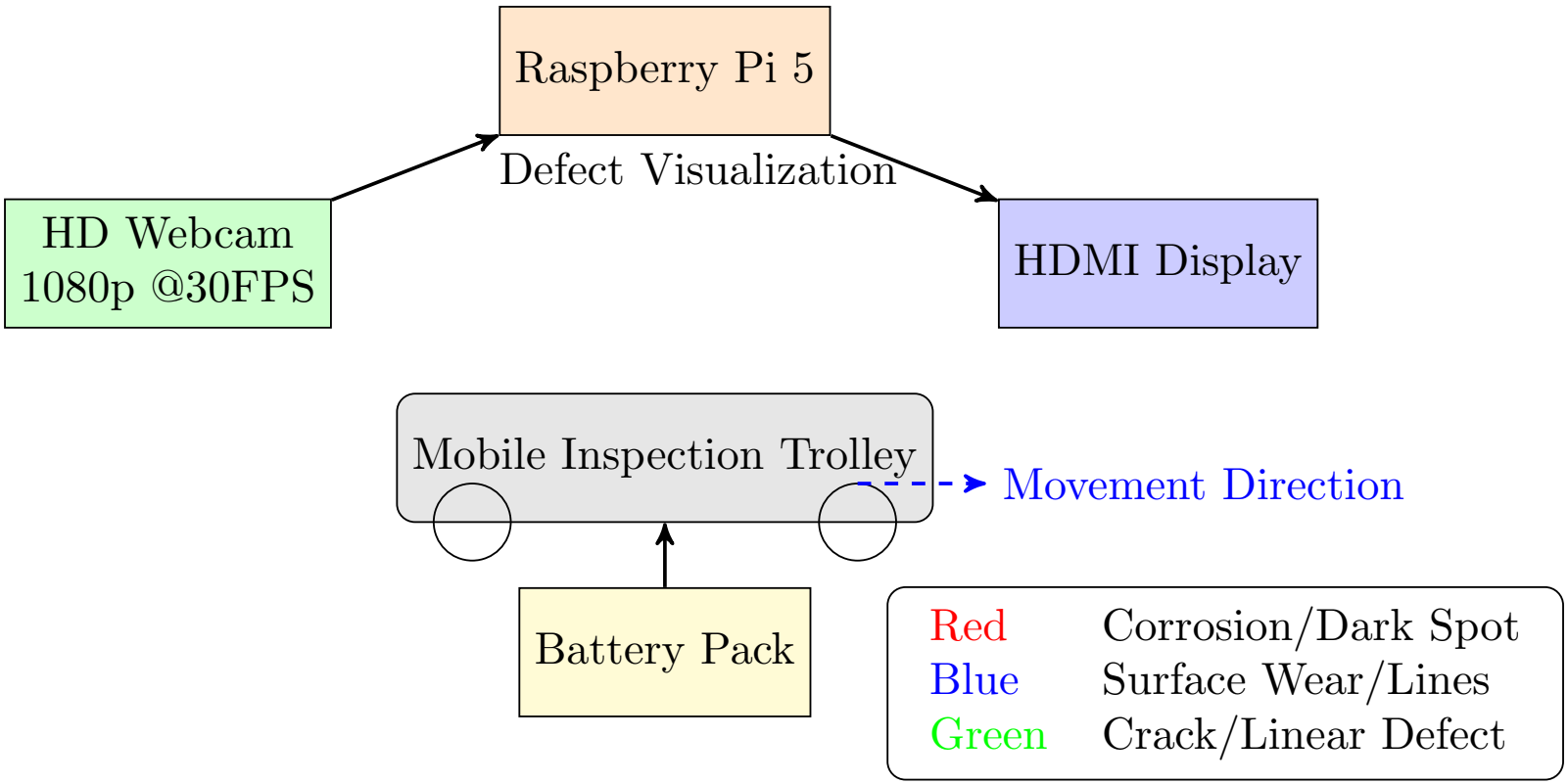
- **Problem**: Manual inspection of third rails is:
  - Time-consuming and labor-intensive
  - Subjective and inconsistent
- **Solution**: Automated image processing system that:
  - Detects defects in real-time
  - Classifies defect types and severity
  - Works for both static images and live video
  - Generates comprehensive reports
- **Business Value**:
  - Improved safety and reliability
  - Reduced maintenance costs
  - Data-driven decision making
- **Status**:
  - Fully functional prototype completed

# Key Achievements

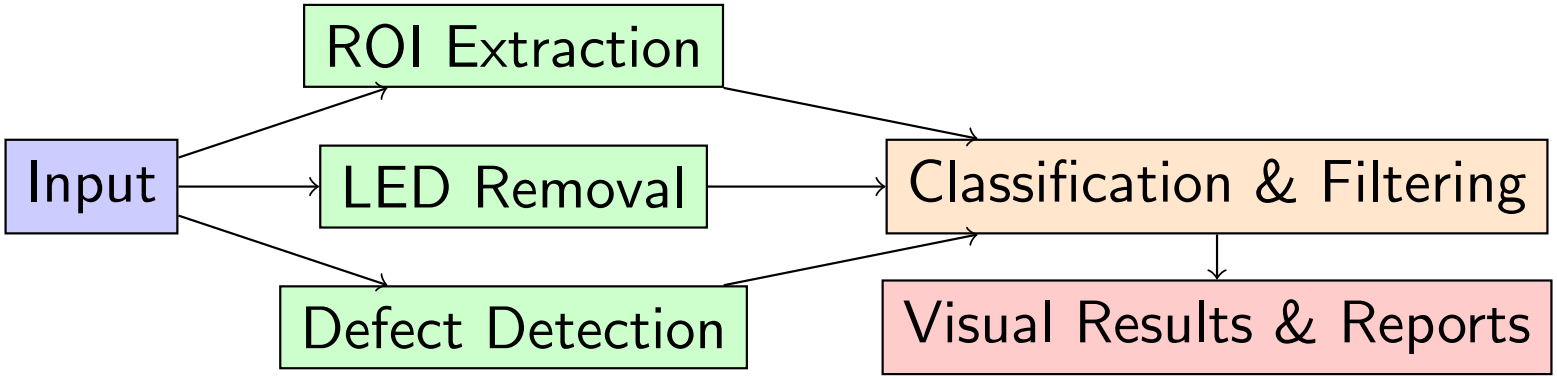
## Delivered Solution

- ✔ Complete automated defect detection system
- ✔ Real-time processing capabilities
- ✔ Multiple defect type classification
- ✔ User-friendly interface with live monitoring

# System Block Diagram



# System Architecture




- `OpenCV` for image processing
- `Python` for system implementation
- Advanced computer vision algorithms
- Real-time processing optimization

# Defect Detection Capabilities

## Corrosion/Dark Spots (Red)

- Rust and oxidation detection
- Severity classification (Low/Medium/High)
- Area-based analysis

## Surface Wear/Lines (Blue)

- Contact wear patterns
- Linear defect identification
- Texture analysis

## Cracks/Linear Defects (Green)

- Structural crack/spot detection
- Length-based severity assessment
- Edge detection algorithms

## Key Features

- Multi-threshold analysis
- Morphological filtering
- Automated classification

# Advanced Image Processing Pipeline

1. **Region of Interest (ROI) Extraction**
   - Focus on critical rail areas (40% width coverage)
   - Reduces processing time and false positives
2. **LED Light Interference Removal**
   - Automatic bright spot detection
   - Intelligent inpainting for glare removal
3. **Multi-Algorithm Defect Detection**
   - Bilateral filtering for noise reduction
   - Adaptive thresholding for varying conditions
   - Hough line detection for crack identification
4. **Intelligent Classification & Filtering**
   - Area and aspect ratio analysis
   - Severity level assignment
   - False positive elimination

# Dual Operating Modes

## Batch Processing Mode

- Process multiple images automatically
- Generate detailed comparison reports
- Suitable for periodic inspections
- Comprehensive defect analysis

## Live Monitoring Mode

- Real-time camera feed processing
- Instant defect visualization
- Recording capabilities (20-second clips)
- Manual capture functionality

# User Interface & Visualization

- **Color-coded defect identification**
  - Red: Corrosion/Dark spots
  - Blue: Surface wear/Lines
  - Green: Cracks/Linear defects
- **Real-time statistics display**
- **Side-by-side comparisons**
- **Interactive controls**
  - Record button
  - Capture button
  - Exit controls

## Output Features

- Severity classification
- Defect count summary
- ROI boundary visualization
- Timestamp annotations
- Automated file naming

# Configurable Parameters

**Detection Thresholds:**

- Corrosion threshold: 50
- Crack threshold: 5
- LED light threshold: 400
- Min/Max defect areas

**Image Processing:**

- Bilateral filter parameters
- Gaussian blur settings
- Canny edge detection
- Morphological operations

**Classification Criteria:**

- Severity thresholds
- Area-based classification
- Aspect ratio analysis
- Length measurements

**Performance Settings:**

- Live frame dimensions
- Recording parameters
- Processing FPS
- Output quality

# System Performance Metrics

## Processing Speed

- Real-time processing
- Batch processing: 2-3 seconds per image

## Detection Accuracy

- High sensitivity to defects
- Low false positive rate
- Multiple validation algorithms

## Resource Efficiency

- Minimal hardware requirements
- Efficient memory usage
- CPU-optimized algorithms

## Reliability Features

- Automatic LED glare removal
- Environmental condition adaptation

# Output Examples & Reporting

## Automated Report Generation

- Side-by-side comparison images
- Defect count summaries with color coding
- Severity level classification
- Timestamped results
- Organized file structure

# Step 0-1: ROI Extraction

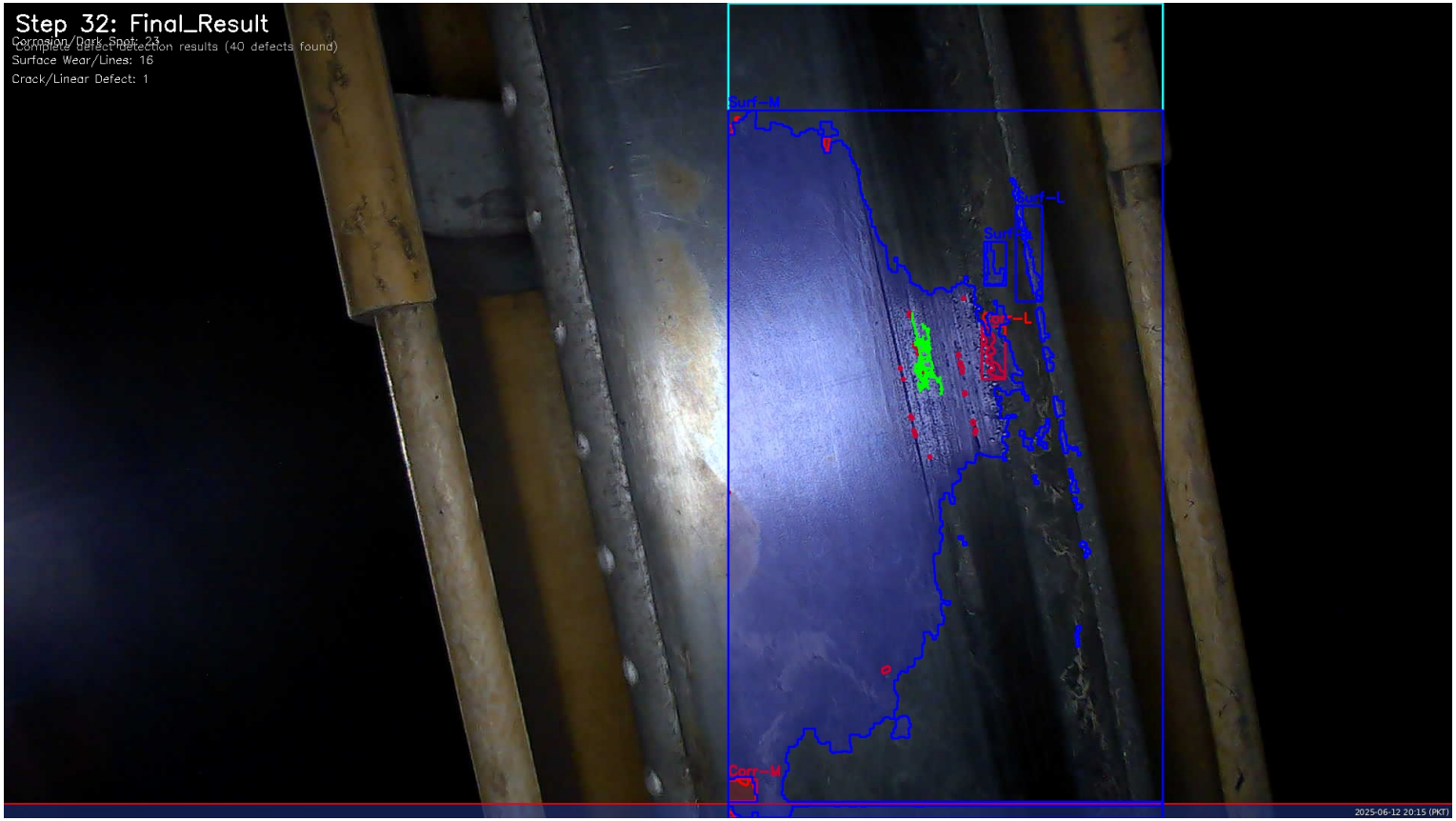
# Step 2: LED Glare Removal

# Step 4: Corrosion Detection

# Step 5: Surface Wear Detection

# Step 6: Crack Detection

# Step 7: Final Classification




- All detected defects labeled with type and severity.

# Step 8: Side by side comparison

- All detected defects labeled with type and severity.

# Output Examples & Reporting

## Automated Report Generation

- Side-by-side comparison images
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# Hardware Requirements

## Minimum Setup

- Raspberry Pi 4 (4GB RAM)
- HD USB/IP camera
- MicroSD card (32GB+)
- LED lighting system
- Network connectivity (optional)

## Software Stack

- Python 3.8+
- OpenCV 4.x
- NumPy, SciPy
- Custom detection algorithms

## Total Investment

- Hardware: 100k PKR
- Software: Developed in-house
- Installation: Minimal

# Training & Support

## Documentation Provided

- Complete user manual
- System operation procedures
- Technical documentation
- Configuration guides

# Key Takeaways

## Mission Accomplished

- ✔ Fully functional defect detection system delivered
- ✔ Exceeds initial requirements and expectations
- ✔ Ready for immediate deployment
- ✔ Significant ROI and safety improvements

## Strategic Advantages

- First-mover advantage in automated rail inspection
- Scalable solution for entire network
- Foundation for future smart infrastructure
- Enhanced reputation for innovation and safety

# Thank You

Questions & Discussion

**Third Rail Defect Detection System**

*Enhancing Railway Safety Through Innovation*